

Project 5 - Prover Infrastructure

- Update
 - (EF DevOps / PandaOps) DIY build hardware preparation - 4x4 cluster (16x 5090 total)
 - zkVM integration (Ere)
 - Add ZisK cluster proving support
 - Update SP1 to v6
 - Publish prover image with CUDA enabled
- Drafted [proof node API](#) for zkAttestor to request for execution proofs (feedback appreciated)
 - Request proof with NewPayloadRequest in SSZ and proof types (async)
 - SSE for proof result event (complete / failure / timeout)
 - Endpoint to get the proof (parameterized by NewPayloadRequest root and proof type)

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- Next
 - Focus on cluster integration
 - Implement proof node API for zkboost (WIP) and integrate with CL
 - Experiment the cluster proving on the 4x4 cluster once it is ready

DIY Build - Per Server Specification

Component	Specification
GPU	4x ASUS TUF RTX 5090 OC (32GB GDDR7, 3.5 slots)
CPU	AMD Threadripper PRO 9975WX (32C/64T, 4GHz base, 5.4GHz boost)
RAM	6x 32GB DDR5 5600 ECC (192GB total)
Motherboard	ASRock WRX90 WS EVO (6x PCIe 5.0 x16, 8 DIMM)
Storage	2x Samsung 9100 PRO 2TB NVMe (RAID 1)
NIC	HPE 841QSFP28 (2x 100G QSFP28, Mellanox ConnectX-5)
Chassis + PSU	OCG4780-4H5-H (4U, dual 3200W PSU)